ROB 501 - MATHEMATICS FOR ROBOTICS

HW #4 Problem Set Solutions

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Ann Arbor, MI
Fall 2018

Problem #1

(a) $\mathscr{S} = \{p_0, p_1, p_2\}$, where $p_0 = 1$, $p_1 = x$ and $p_2 = x^2$. Hence,

$$r(x) = 2 + 3x - x^2 = c_0 p_0 + c_1 p_1 + c_2 p_2 \Longrightarrow \boxed{c_0 = 2}, \boxed{c_1 = 3} \text{ and } \boxed{c_2 = -1}.$$

(b) $r(x) = d_0 q_0 + d_1 q_1 + d_2 q_2 = d_0(1) + d_1(1 - x) + d_2(x + x^2)$. Hence, $\boxed{d_0 = 6}$, $\boxed{d_1 = -4}$ and $\boxed{d_2 = -1}$.

The change of basis matrix from $\mathscr{Q}$ to $\mathscr{S}$ is easy to determine by appending the representations of p_0, p_1 and p_2 in the $\mathscr{Q}$ basis. Thus,

$$[p_0]_{\mathscr{Q}} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \ [p_1]_{\mathscr{Q}} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \text{ and } [p_2]_{\mathscr{Q}} = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \Longrightarrow \boxed{P = \begin{bmatrix} 1 & 1 & -1 \\ 0 & -1 & 1 \\ 0 & 0 & 1 \end{bmatrix}}.$$

Problem #2

The e-values for tha matrix A_3 are: $\lambda_1 = 1$, $\lambda_2 = 2$ and $\lambda_3 = 3$.

To calculate the e-vector v^1 associated to λ_1, we can choose its first component as 1 and find the other components. Thus,

$$\begin{bmatrix} 1 & 4 & 10 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ y_1 \\ z_1 \end{bmatrix} = \lambda_1 \begin{bmatrix} 1 \\ y_1 \\ z_1 \end{bmatrix} \Longrightarrow \begin{cases} 1+4y_1 \quad +10z_1 = 1 \\ \quad 2y_1 \qquad\quad = y_1 \\ \qquad\qquad 3z_1 = z_1 \end{cases} \Longrightarrow \boxed{v^1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}}.$$

To calculate v^2 associated to λ_2 and following the same reasoning, we have

$$\begin{bmatrix} 1 & 4 & 10 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ y_2 \\ z_2 \end{bmatrix} = \lambda_2 \begin{bmatrix} 1 \\ y_2 \\ z_2 \end{bmatrix} \Longrightarrow \begin{cases} 1+4y_2 \quad +10z_2 = 2 \\ \quad 2y_2 \qquad\quad = 2y_2 \\ \qquad\qquad 3z_2 = 2z_2 \end{cases} \Longrightarrow \boxed{v^2 = \begin{bmatrix} 1 \\ \frac{1}{4} \\ 0 \end{bmatrix}}.$$

Finally, to calculate v^3 associated to λ_3, we have

$$\begin{bmatrix} 1 & 4 & 10 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ y_3 \\ z_3 \end{bmatrix} = \lambda_3 \begin{bmatrix} 1 \\ y_3 \\ z_3 \end{bmatrix} \Longrightarrow \begin{cases} 1+4y_3 \quad +10z_3 = 3 \\ \quad 2y_3 \qquad\quad = 3y_3 \\ \qquad\qquad 3z_3 = 3z_3 \end{cases} \Longrightarrow \boxed{v^3 = \begin{bmatrix} 1 \\ 0 \\ \frac{1}{5} \end{bmatrix}}.$$

To verify that the vectors are linearly independent, we can check if the determinant of the matrix $[v^1 \,|\, v^2 \,|\, v^3]$ is 0:

$$\det \begin{bmatrix} 1 & 1 & 1 \\ 0 & \frac{1}{4} & 0 \\ 0 & 0 & \frac{1}{5} \end{bmatrix} = \frac{1}{20} \neq 0 \Longrightarrow \text{L.I.}$$

The e-vectors calculated by using the `eig` command in MATLAB are:

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0.9701 \\ 0.2425 \\ 0 \end{bmatrix} \text{ and } \begin{bmatrix} 0.9806 \\ 0 \\ 0.1961 \end{bmatrix}.$$

Only the first e-vector is the same, but the other two are parallel to v^2 and v^3. It happens because MATLAB's e-vectors are unit vectors, i.e., they have length equal to 1.

Problem #3

To compute the e-values, we have to find the roots of the characteristic polynomial:

$$\det\left(\lambda I - A_4\right) = \begin{vmatrix} (\lambda-3) & -1 & 0 \\ 0 & (\lambda-3) & 0 \\ 0 & 0 & (\lambda-2) \end{vmatrix} = (\lambda-3)^2(\lambda-2) \Longrightarrow \begin{cases} \lambda_1 = \lambda_2 = 3 \\ \lambda_3 = 2 \end{cases} .$$

It is possible to calculate the e-vector v^3 associated to the e-value $\lambda_3 = 2$, because its multiplicity is 1. In this case, $v^3 = [0\ 0\ 1]^\top$ is a possible choice. However, when we try to calculate the e-vectors associated to λ_1 or λ_2, which are both equal to 3, it is possible to find only vectors parallel to $v^1 = [1\ 0\ 0]^\top$. Therefore, we can only have two linearly indepedent e-vectors and they can't be a basis for $\mathbb{R}^3$.

Problem #4

A and B are similar, then $B = P^{-1}AP$. The characteristic polynomial of A is $\det(\lambda I - A)$.

Pre-multiplying $(\lambda I - A)$ by P^{-1} and pos-multiplying by P, we have,

$$P^{-1}(\lambda I - A)P = (\lambda P^{-1}IP - P^{-1}AP) = (\lambda I - B) \; .$$

Hence, applying the determinant to both sides and remembering that $\det(P^{-1}) = \frac{1}{\det(P)}$, we have

$$\det(\lambda I - B) = \det\left[P^{-1}(\lambda I - A)P\right] = \cancel{\det(P^{-1})}\det(\lambda I - A)\cancel{\det(P)} = \det(\lambda I - A) \; .$$

Therefore, they have the same characteristic polynomial and, consequently, the same characteristic equation. $\square$

Problem #5

Let $P = [v^1|v^2|v^3]$ be the matrix obtained by appending the eigenvectors of the matrix A_3. Multiplying A_3 and P we have

$$A_3P = [A_3v^1|A_3v^2|A_3v^3] = [\lambda_1 v^1|\lambda_2 v^2|\lambda_3 v^3] = [v^1|v^2|v^3] \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} = P\Lambda$$

The eigenvectors are linearly indepedent, and therefore $\det(P) \neq 0$. Hence, P is invertible and we can pre-multiply both sides of the equation above by P^{-1}, coming to

$$P^{-1}A_3P = \Lambda \ .$$

Therefore, A_3 and Λ are similar matrices.[1] $\square$

[1]In this specific case, we have $\Lambda = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, $P = \begin{bmatrix} 1 & 1 & 1 \\ 0 & \frac{1}{4} & 0 \\ 0 & 0 & \frac{1}{5} \end{bmatrix}$ and $P^{-1} = \begin{bmatrix} 1 & -4 & -5 \\ 0 & 4 & 0 \\ 0 & 0 & 5 \end{bmatrix}$.

Problem #6

(a) To show that L is a linear operator, we must show that $\forall\,\alpha,\beta\in\mathbb{R}$, and $\forall\,M_1,M_2\in\mathbb{R}^{2\times2}$. We have

$$L(\alpha M_1+\beta M_2)=\alpha L(M_1)+\beta L(M_2)\ .$$

Thus,

$$L(\alpha M_1+\beta M_2)=\frac{1}{2}[(\alpha M_1+\beta M_2)+(\alpha M_1+\beta M_2)^\top]=$$
$$\frac{1}{2}[(\alpha M_1)+(\alpha M_1)^\top]+\frac{1}{2}[(\beta M_2)+(\beta M_2)^\top]=\alpha\frac{1}{2}[(M_1)+(M_1)^\top]+\beta\frac{1}{2}[(M_2)+(M_2)^\top]=$$
$$=\alpha L(M_1)+\beta L(M_2)\ .\ \square$$

(b) At first, we should represent the basis as vectors:

$$E^{11}=\begin{bmatrix}1\\0\\0\\0\end{bmatrix},\ E^{12}=\begin{bmatrix}0\\1\\0\\0\end{bmatrix},\ E^{21}=\begin{bmatrix}0\\0\\1\\0\end{bmatrix}\text{ and }E^{22}=\begin{bmatrix}0\\0\\0\\1\end{bmatrix}.$$

Now, we should apply the linear operator and then represent the results as vectors:

$$L(E^{11})=\tfrac{1}{2}\left[\begin{bmatrix}1&0\\0&0\end{bmatrix}+\begin{bmatrix}1&0\\0&0\end{bmatrix}^\top\right]=\begin{bmatrix}1&0\\0&0\end{bmatrix}=E^{11}=\begin{bmatrix}1\\0\\0\\0\end{bmatrix};$$

$$L(E^{12})=\tfrac{1}{2}\left[\begin{bmatrix}0&1\\0&0\end{bmatrix}+\begin{bmatrix}0&1\\0&0\end{bmatrix}^\top\right]=\begin{bmatrix}0&\frac{1}{2}\\\frac{1}{2}&0\end{bmatrix}=\tfrac{1}{2}E^{12}+\tfrac{1}{2}E^{21}=\begin{bmatrix}0\\\frac{1}{2}\\\frac{1}{2}\\0\end{bmatrix};$$

$$L(E^{21})=\tfrac{1}{2}\left[\begin{bmatrix}0&0\\1&0\end{bmatrix}+\begin{bmatrix}0&0\\1&0\end{bmatrix}^\top\right]=\begin{bmatrix}0&\frac{1}{2}\\\frac{1}{2}&0\end{bmatrix}=\tfrac{1}{2}E^{21}+\tfrac{1}{2}E^{12}=\begin{bmatrix}0\\\frac{1}{2}\\\frac{1}{2}\\0\end{bmatrix};$$

$$L(E^{22})=\tfrac{1}{2}\left[\begin{bmatrix}0&0\\0&1\end{bmatrix}+\begin{bmatrix}0&0\\0&1\end{bmatrix}^\top\right]=\begin{bmatrix}0&0\\0&1\end{bmatrix}=E^{22}=\begin{bmatrix}0\\0\\0\\1\end{bmatrix}.$$

Appending the resultant vectors, we have

$$\boxed{A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}.$$

Problem #7

(a) As $\mathbb{C}$ is the field, the canonical basis for $\mathbb{C}^n$ is given by $\{e_1, e_2, \dots, e_n\}$ where the vectors can be represented by

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \; e_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \; \dots, e_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}.$$

Applying the linear transformation to the vectors of the basis we have

$$L(e_1) = A \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = A^1, \; L(e_2) = A \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix} = A^2, \; \dots, L(e_n) = A \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix} = A^n,$$

where A^i is the i-th columns of the matrix A. Thus,

$$\hat{A} = \left[A^1 | A^2 | \dots | A^n \right] = A$$

(b) Let $\{v^1, v^2, \dots, v^n\}$ be the basis constructed from the eigenvectors of A and $\{\lambda_1, \lambda_2, \dots, \lambda_n\}$ the respective associated eigenvalues. Applying the linear transformation to the vectors representation on this the basis, we have

$$L(v^1) = Av^1 = \lambda_1 v^1 \Longrightarrow \hat{A} \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} \lambda_1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \; \dots, L(v^n) = Av^n = \lambda_n v^n \Longrightarrow \hat{A} \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ \lambda_n \end{bmatrix}.$$

The columns of $\hat{A}$ are the results of the operations above. Thus,

$$\hat{A} = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix}$$

Problem #8

(a) By using the commands provided, we get Figure 1.

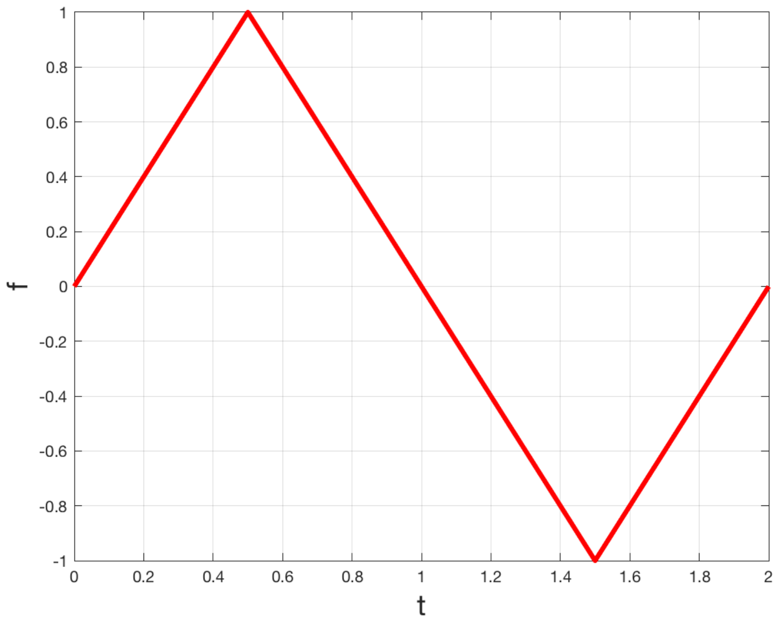

Figure 1: Triangle Wave.

(b) The coefficients obtained are

$$
\begin{aligned}
c_0 &= -0.0370; \\
c_1 &= 2.2158; \\
c_2 &= 2.8901; \\
c_3 &= -11.2271; \\
c_4 &= 7.6978; \\
c_5 &= -1.5396\,.
\end{aligned}
$$

The plot is shown in Figure 2.

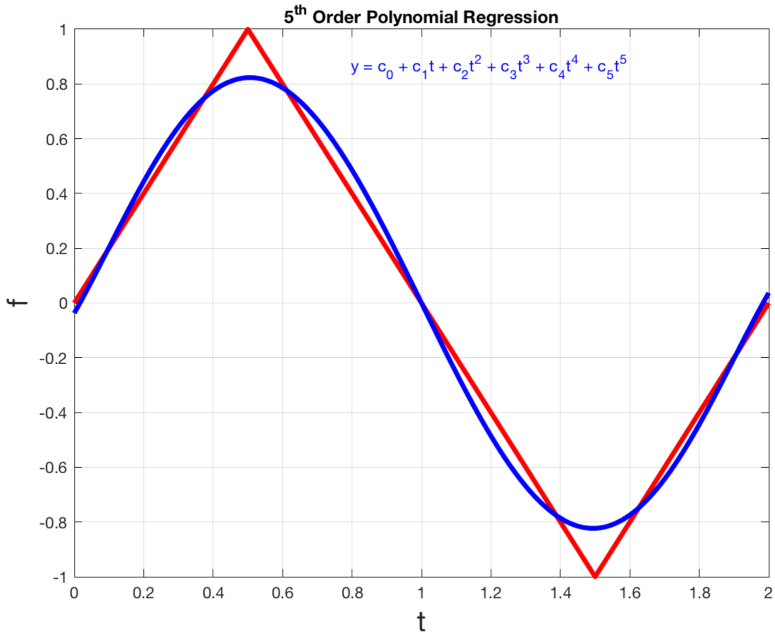

Figure 2: Triangle Wave and 5-th order polynomial fit.

(c) The coefficients obtained are

$$\begin{aligned} c_1 &= 0.8106; \\ c_2 &= 0.00; \\ c_3 &= -0.0901; \\ c_4 &= 0.00; \\ c_5 &= 0.0325\,. \end{aligned}$$

The plot is shown in Figure 3.

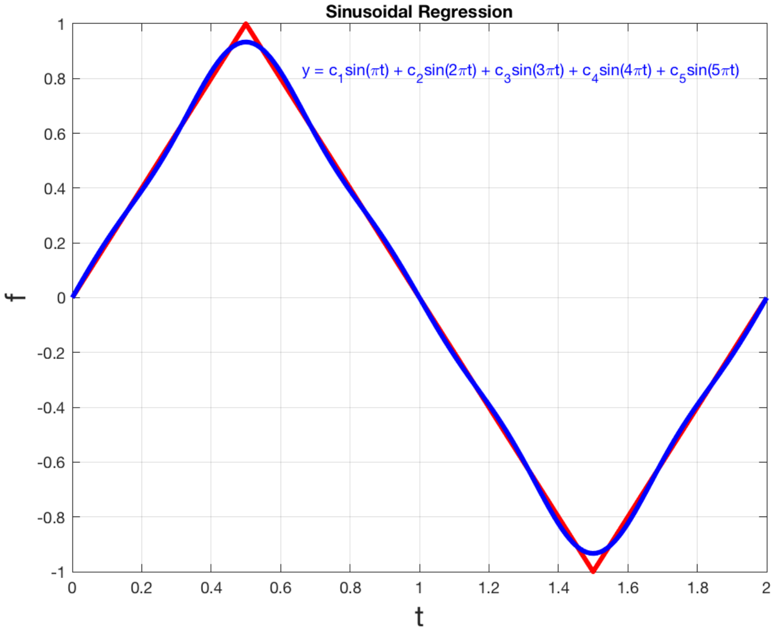

Figure 3: Triangle Wave and sinusoidal fit.

Problem #9

Using the basis $\{1, t, t^2, t^3, t^4, t^5\}$, the coefficients obtained are

$$\begin{aligned}
c_0 &= 0.0428;\\
c_1 &= 0.0505;\\
c_2 &= -2.4087;\\
c_3 &= 3.6850;\\
c_4 &= -0.2325;\\
c_5 &= -0.1186\,.
\end{aligned}$$

The derivative of $f(t)$ in $t = 0.3$ is -0.4297. The plot is shown in Figure 4.

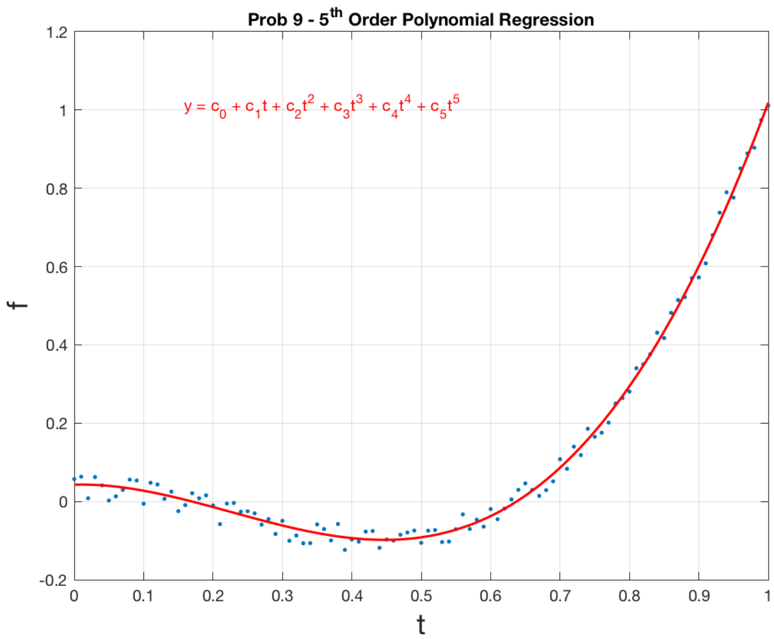

Figure 4: Noisy data and 5-th order polynomial fit.

Using the basis $\{1, e^t, e^{2t}, e^{3t}, e^{4t}, e^{5t}\}$, the coefficients obtained are

$$\begin{aligned} c_0 &= -4.4237; \\ c_1 &= 13.9932; \\ c_2 &= -16.0732; \\ c_3 &= 8.3828; \\ c_4 &= -2.0186; \\ c_5 &= 0.1887\,. \end{aligned}$$

The derivative of $f(t)$ in $t = 0.3$ is -0.4096. The plot is shown in Figure 5.

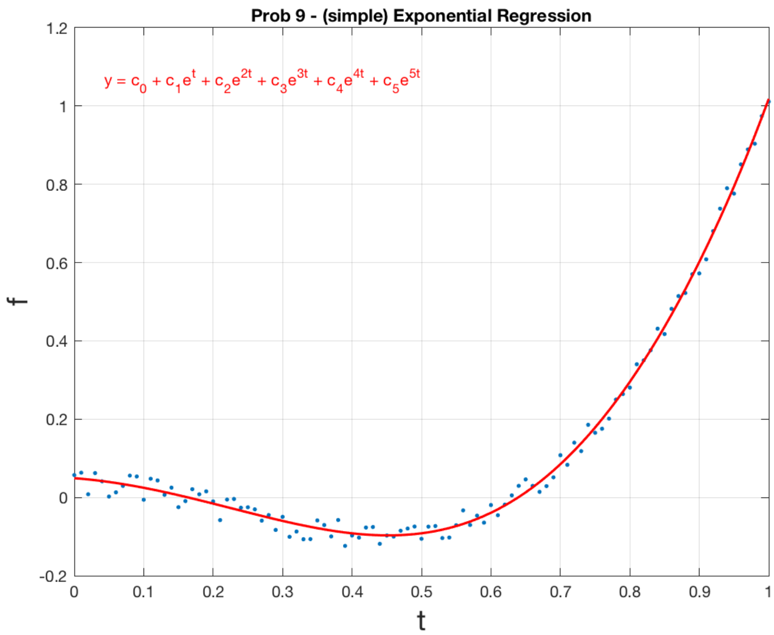

Figure 5: Noisy data and exponential fit.

Finally, using the basis $\{1, e^t, e^{t^2}, e^{t^3}, e^{t^4}, e^{t^5}\}$, the coefficients obtained are

$$\begin{aligned} c_0 &= -0.5304; \\ c_1 &= 0.3636; \\ c_2 &= -5.5606; \\ c_3 &= 13.8488; \\ c_4 &= -11.2299; \\ c_5 &= 3.1445\,. \end{aligned}$$

The derivative of $f(t)$ in $t = 0.3$ is -0.4132. The plot is shown in Figure 6.

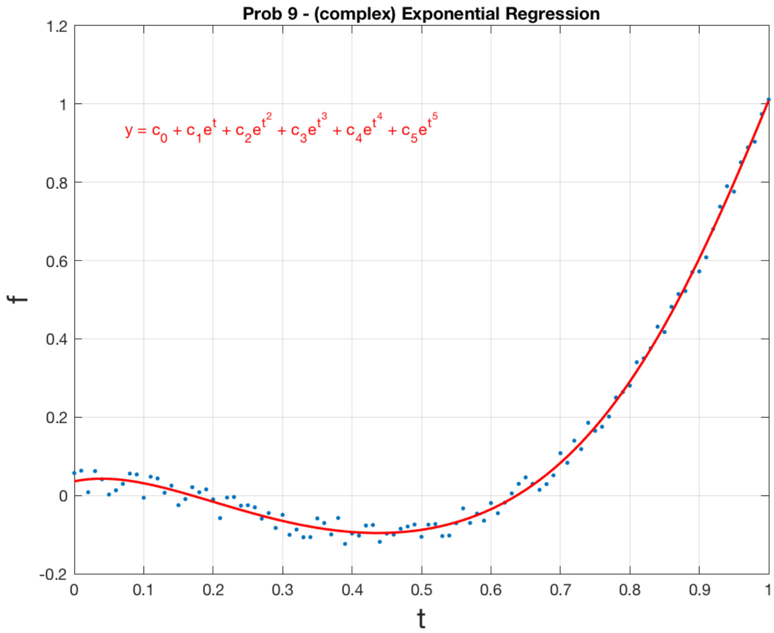

Figure 6: Noisy data and exponential of monomials fit.